

MIPS & RISC-V Instruction Set – Concise Exam Reference (CT-1)

MIPS Instruction Set (Core)

Category	Instructions
Arithmetic	add, addu, sub, subu, addi, addiu
Logical	and, or, xor, nor, andi, ori, xori
Shift	sll, srl, sra
Comparison	slt, slti
Memory	lw, sw, lb, sb
Branch	beq, bne
Jump	j, jal, jr

RISC-V Instruction Set (RV32I Core)

Category	Instructions
Arithmetic	add, sub, addi
Logical	and, or, xor, andi, ori, xori
Shift	sll, srl, sra, slli, srli, srai
Comparison	slt, slti, sltu, sltiu
Memory	lw, sw, lb, sb, lh, sh
Branch	beq, bne, blt, bge
Jump	jal, jalr
Upper Imm	lui, auipc

Exam Tip: For CT-1, focus on load-store instructions, arithmetic ops, branches, and how they interact in the 5-stage pipeline. Instruction behavior matters more than encoding.